

Basic HTML document structure

SECTION 1 - What Is This? (Plain English, Zero Jargon)

Imagine you are sending a formal letter through the post office. For that letter to reach its destination and be understood, it needs a specific setup. You don't just write words on a scrap of paper and throw it in a mailbox.

First, there is the **envelope** (the `<html>` tag), which holds everything together and tells the post office, "This is a letter." On the outside of the envelope, you have the **delivery instructions** and metadata—the address, the stamp, and the return info (the `<head>` tag). These instructions are for the post office, not for the person reading the letter. Finally, inside the envelope, you have the **actual paper** with your message on it (the `<body>` tag). This is what the recipient actually sees and reads.

In web development, every webpage follows this exact "envelope" system. It ensures that the browser (the post office) knows how to handle the file before it even starts showing the content to the user. Without this structure, the browser might get confused about which part is a message and which part is just background information.

SECTION 2 - The Very First Step (Absolute Basics, From Scratch)

We are going to build the "skeleton" that exists behind every single website on the internet.

1. Create the File

- Open your text editor.
- Create a new file and save it as `structure.html`.

2. Write the Skeleton

Type these four essential parts into your file:

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
</head>
```

```
<body>
```

This is where the content goes!

```
</body>
```

```
</html>
```

3. How to Run It

- **Command:** Open your file explorer, find `structure.html`, and double-click it.
 - **Expected Output:** You will see a blank white page with the text "This is where the content goes!" in the top left corner. Even though the page looks simple, the browser has successfully read the invisible "envelope" instructions.
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SECTION 3 - Technical Explanation (Proper Terms + How It Works Internally)

While the previous session taught us what HTML is, this session focuses on the **Document Skeleton**. Here is what the browser's engine is doing with each line:

- **<!DOCTYPE html>**: This is a **document type declaration**. It isn't an HTML tag; it's an instruction to the browser that this is an **HTML5** document. Internally, this prevents the browser from entering "Quirks Mode," where it tries to emulate bugs from 1990s browsers.
- **<html>**: This is the **root element**. Every other tag must be nested inside this. It defines the boundary of the web document.
- **<head>**: This section contains **metadata**—data about data. The browser processes this first to set the character encoding, the page title, and link external files. None of the content inside **<head>** is painted on the screen.
- **<body>**: This is the **viewable area**. Anything placed here is parsed and rendered into the **DOM (Document Object Model)** tree to be displayed to the user.

Common Gotcha: Putting content tags (like `<h1>` or `<p>`) inside the **<head>** by mistake. While browsers might try to fix this, it can lead to "Flash of Unstyled Content" or broken SEO because search engines expect the **<head>** to contain only metadata.

SECTION 4 - Why We Use This

1. **What specific problem does this solve?** Standardization. Without a formal structure, different browsers (Chrome vs. Safari) would interpret where a page starts and ends differently, leading to inconsistent layouts.
2. **When SHOULD you use this?** Always. Even the smallest web project requires this basic skeleton to be considered a valid HTML document.
3. **When is it overkill?** It is never overkill for a web page. However, if you are writing a small "component" or "snippet" of code to be inserted into an existing page (like a React component), you don't include the `<html>` or `<body>` tags because the main page already has them.

SECTION 5 - Alternatives & Comparisons

(Interview-Ready)

Alternative	What is it?	How it differs
Older DOCTYPEs	Long, complex declarations for HTML 4.01 or XHTML.	HTML5 simplified this to just <code><!DOCTYPE html></code> . Older versions required a URL link to a "DTD" file to define the rules.
Single Page Apps (SPA)	Frameworks like React or Vue.	In an SPA, you often have only one HTML file with this structure. All other "pages" are just swapped into the <code><body></code> using JavaScript without reloading the skeleton.
Fragment Files	Files like <code>.php</code> or <code>.ejs</code> partials.	These are "pieces" of a page. They lack the <code><html></code> and <code><body></code> tags because they are meant to be "glued" together by a server before being sent to the browser.

SECTION 6 - Code Example (Complete, Commented, Runnable)

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Skeleton Demo</title>

</head>


<body>

  <h1>The HTML Skeleton</h1>

  <p>Every professional website starts with these few lines of code.</p>

</body>


</html>
```

SECTION 7 - Common Interview Questions

1. **What is the purpose of the `lang="en"` attribute in the `<html>` tag?** It identifies the primary language of the document. This is crucial for **accessibility**, as it tells screen readers which language to use for text-to-speech, and for SEO, as it helps search engines categorize the site.
2. **Why is `<!DOCTYPE html>` not considered an HTML tag?** It is a **declaration** or a "prologue." It doesn't have a closing tag and it doesn't represent content; it purely informs the browser's parser about the version of HTML being used.
3. **Can a webpage have more than one `<body>` tag?** No. A valid HTML document must have exactly one `<body>` tag. Having multiple can confuse the browser and lead to unpredictable rendering of the DOM tree.
4. **What is the difference between `<head>` and `<header>`?** The `<head>` is for **metadata** and instructions that don't appear on the page. The `<header>` (which we will learn later) is a visible section used for things like navigation bars and logos within the `<body>`.

5. **What happens if you leave out the `<html>` tags?** Modern browsers are very forgiving and will likely "auto-correct" the mistake by wrapping your code in an invisible `<html>` element. However, this is bad practice as it can break some JavaScript functions and CSS styles that rely on a proper root element.

Interview summary: If asked to explain Basic HTML document structure in one sentence, say: "The basic HTML structure consists of the DOCTYPE declaration followed by nested html, head, and body tags, which together separate the document's metadata from its viewable content."

SECTION 8 - Mini-Project (Build It Right Now)

What to build: A "Code Checklist" page that lists and explains the five essential parts of the skeleton.

Steps:

1. **Create a file** named `checklist.html`.
2. **Inside the file**, write out a perfect HTML5 skeleton from memory.
3. **In the `<title>`**, put "HTML Skeleton Checklist."
4. **In the `<body>`**, use an `<h1>` for the main title: "The 5 Pillars of HTML."
5. **Below the title**, write five separate `<p>` tags. Each paragraph should name one part (DOCTYPE, html, head, title, body) and one short sentence explaining it in your own words.
6. **Expected Output:** A clean, valid HTML page that serves as your own personal "cheat sheet" for future sessions.

Stretch Goal: Add a `<meta>` tag for "description" in the `<head>` that summarizes what your checklist is about (Hint: Search for "meta description tag").

Interview summary: If asked to explain FE-02 Basic HTML document structure in one sentence, say: "The basic HTML structure consists of the DOCTYPE declaration followed by nested html, head, and body tags, which together separate the document's metadata from its viewable content."